

# Profile-Driven Hardware Simulation for Organic Optoelectronic Edge Vision

Songqiao Li

## Abstract

We present a behavioral simulation framework for organic optoelectronic inference mapping literature metrics to task-level accuracy. Across three datasets, ADC resolution dominates under the canonical regime: below 6 bits causes abrupt collapse, while above 6 bits device-to-device variability becomes binding. Standard hardware-aware training overfits a single fixed mismatch and collapses to 10.00% on fresh hardware, whereas Ensemble HAT—resampling mismatch masks each epoch—recovers  $86.37 \pm 1.54\%$ . A literature-anchored OPECT case study reaches  $88.53 \pm 0.08\%$  zero-shot transfer. Under severe nonlinear write ( $NL=2.0$ ), we systematically test and falsify three candidate mitigation strategies, including joint training, which remains pinned at  $\sim 30\%$  fresh-instance accuracy. Our findings establish a  $\sim 30\%$  fresh-instance ceiling under severe nonlinearity, indicating that first-order behavioral surrogates impose a structural generalization barrier in the attention pathway.

**Keywords:** organic optoelectronic compute-in-memory; hardware-aware training; vision transformers; edge vision; mixed-signal inference; profile-driven simulation.

## 1 Introduction

Compute-in-memory (CIM) architectures execute dense matrix-vector multiplications directly within memory arrays, eliminating the data-movement bottleneck that limits conventional digital systems<sup>1-3</sup>. Beyond conventional static random-access memory (SRAM) and emerging resistive memories, organic optoelectronic devices offer multilevel tunability, low-voltage operation, and inherent photosensitivity for direct optical signal processing<sup>4-6</sup>. Recent demonstrations include analog conductance states<sup>5,7</sup>, long retention tails<sup>8</sup>, and integrated optoelectronic synaptic arrays<sup>9-11</sup>.

However, translating these device-level advances into system-level vision performance faces a critical methodological gap. The organic optoelectronic literature remains dominated by device-centric characterizations<sup>12</sup>. Network-level studies are typically limited to small benchmarks or simple perceptrons, while variability, ADC resolution, retention-induced drift, and transfer across fabricated instances are either absent from system analyses or treated only qualitatively<sup>7,8</sup>. The most relevant precursor for variability-aware organic simulation remains the Monte Carlo study of Lin *et al.*, which sampled device parameters for a simple organic-memristive perceptron<sup>13</sup>. No clear framework yet connects reported organic-device characteristics to deployment-facing accuracy expectations for modern Vision Transformer (ViT) or convolutional neural network (CNN) backbones.

This gap is particularly acute for hardware-aware training (HAT), which mitigates analog non-idealities by injecting device noise into the forward training path<sup>14</sup>. Although related in spirit to quantization-aware training<sup>15</sup> and to domain randomization in sim-to-real transfer<sup>16</sup>, HAT for CIM faces an additional difficulty: device-to-device (D2D) mismatch is spatially structured and fixed for each hardware instance, unlike independent thermal noise. Standard HAT therefore tends to optimize against one sampled mismatch map. As we show below, this choice can make the model fit a particular hardware instance rather than the deployment distribution, leading to catastrophic accuracy collapse when transferring to fresh hardware.

System-level CIM simulators for inorganic memories—DNN+NeuroSim<sup>2</sup>, MemTorch<sup>17</sup>, AIHWKIT<sup>18</sup>, and CrossSim<sup>19</sup>—do not directly capture the coupled photoresponse, retention, and write nonlinearity that define organic optoelectronic inference.

Recent organic array demonstrations<sup>9;10;20;21</sup> and system-constraint studies<sup>22–26</sup> provide plausible parameter ranges for conductance windows, retention, and array-level artifacts. At the network level, prior CIM practice motivates a hybrid analog–digital deployment: dense linear projections map naturally to crossbars, whereas dynamic attention and irregular operators remain digital because of precision requirements<sup>27–33</sup>.

Here we present a behavioral simulation framework that bridges this materials-to-system gap, providing structured identification and ranking of hardware-induced failure modes—what we descriptively call risk-aware evaluation—prior to fabrication closure. The framework incorporates organic-device-specific non-idealities—photoresponse, retention, and write nonlinearity—through a replaceable device-profile interface and applies a common mixed-signal deployment model to ResNet-18, ConvNeXt-Tiny, and Tiny-ViT-5M. Four constraints dominate in this setting. First, ADC resolution rather than nominal conductance quantization becomes the primary accuracy bottleneck: reducing ADC resolution below 6 bits causes abrupt collapse. Second, the organic phototransistor front end introduces a sublinear photoresponse ( $I_{\text{photo}} \propto P^{\gamma_{\text{phys}}}$ ) that distorts input intensity; we show that an inverse-gamma preprocessor recovers accuracy for strongly nonlinear devices ( $\gamma_{\text{phys}} = 2.0$ , +5.8 percentage points), but that the same compensation amplifies shot-noise variance at high intensities—a signal-to-noise trade-off that must be balanced at the profile level. Third, standard HAT overfits a fixed D2D realization and fails under fresh-instance transfer; Ensemble HAT, which resamples D2D masks at each training epoch, raises fresh-instance accuracy to  $86.37 \pm 1.54\%$ . Fourth, severe nonlinear write ( $NL = 2.0$ ) remains difficult to recover under the present gradient-scaling approximation, limiting accuracy to  $27.72 \pm 0.82\%$ .

To make the scope explicit, this study makes four concrete contributions. First, it establishes a profile-driven first-order behavioral workflow for organic optoelectronic CIM that maps literature-derived device characteristics to task-level vision accuracy while retaining a common mixed-signal deployment model across convolutional neural network (CNN) and Vision Transformer (ViT) backbones. Second, it identifies the dominant deployment constraints in this regime through a two-stage hierarchy: a sub-6-bit ADC cliff quantified by Sobol analysis ( $S_{\text{ADC}} = 0.98$ ), and a physically-motivated inverse-gamma front-end compensation that restores accuracy for strongly sublinear photoresponse ( $\gamma_{\text{phys}} = 2.0$ , +5.8 pp) while explicitly trading dark-region linearity for bright-region shot-noise amplification. Third, it exposes a fresh-instance transfer failure mode in standard HAT and introduces Ensemble HAT, which resamples D2D masks at each training epoch to raise fresh-instance accuracy from chance level (10.00%) to  $86.37 \pm 1.54\%$ . Fourth, it turns two frequently qualitative concerns—literature-profile substitution and severe nonlinear write—into bounded system-level evidence by demonstrating zero-shot transfer to a literature-anchored OPECT profile and by localizing the present  $NL = 2.0$  surrogate failure primarily to the multi-layer perceptron (MLP) path of the transformer.

The framework is intentionally behavioral rather than circuit-accurate: array-level parasitics (current–resistance, or IR, drop; sneak paths), thermal effects, and pulse-faithful programming dynamics are not modeled explicitly and are revisited in Section 4. Its purpose is to identify which device characteristics most constrain edge-vision accuracy before full array hardware becomes available.

## 2 Related Work

### 2.1 Hardware-Aware Training and Robustness

Hardware-aware training (HAT) mitigates analog non-idealities by injecting hardware perturbations into the forward path during optimization<sup>14</sup>. Although related in spirit to quantization-

aware training<sup>15</sup> and to domain randomization in sim-to-real transfer<sup>16</sup>, HAT for CIM faces an additional difficulty: device-to-device (D2D) mismatch is spatially structured and fixed for a given hardware instance. It is therefore different from independent thermal or sampling noise. Standard HAT usually keeps one random D2D mask fixed throughout training. Section 3.4 shows that this choice causes the model to fit a particular hardware instance rather than the deployment distribution. Ensemble HAT resamples the static D2D mask at each training epoch, improving zero-shot transfer to fresh hardware instances. Existing open-source analog-training stacks are closer to i.i.d. perturbation injection than to full spatial-instance resampling<sup>34</sup>. In particular, AIHWKIT exposes practical analog noise injection and per-batch stochastic variation, but not a paper-locked baseline that resamples complete structured D2D mismatch maps for each hardware instance. Our reviewer-facing comparisons therefore rely on internal controls—fixed-mask HAT, exploratory per-batch perturbation, and epoch-level structured resampling—rather than on a nonexistent apples-to-apples external baseline.

## 2.2 Organic Neuromorphic and Optoelectronic Devices

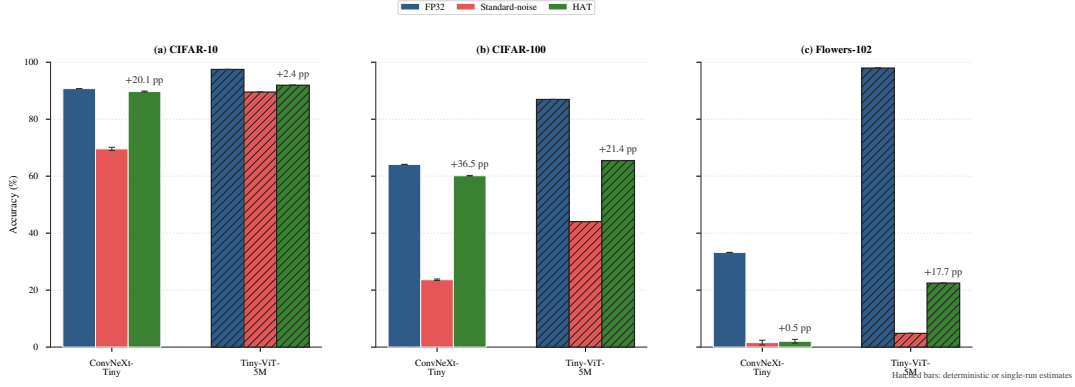
Organic synaptic and memristive devices offer optical sensitivity, multilevel conductance tuning, and flexible-substrate compatibility for neuromorphic hardware<sup>4;6;35;36</sup>. Recent demonstrations include multilevel memory<sup>5;7</sup>, long retention tails<sup>8</sup>, and integrated optoelectronic arrays<sup>9;10;20;21</sup>. Array-level studies are beginning to treat active-matrix addressing, spatial optical response, and crosstalk as system constraints rather than purely materials observations<sup>22–24</sup>, while temperature-resilient organic synapses and high-temperature synaptic phototransistors suggest that thermal stability is also becoming deployment-relevant<sup>25;26</sup>.

At the same time, most of this literature remains device-centric. Network demonstrations are often limited to idealized mappings or small benchmarks, and variability is either absent or treated only qualitatively. The most relevant precursor for variability-aware organic simulation remains the Monte Carlo study of Lin *et al.*, which sampled device parameters for a simple organic-memristive perceptron<sup>13</sup>. The present work follows that variability-aware perspective, but extends it to modern deep-learning software, CIFAR-scale tasks, and more expressive backbones.

## 2.3 CIM Simulation Frameworks and Hybrid Mapping

System-level CIM simulators such as DNN+NeuroSim established an effective template for connecting device assumptions, peripheral-circuit costs, and neural-network accuracy<sup>2</sup>. MemTorch showed how memristive non-idealities can be embedded into PyTorch workflows for end-to-end evaluation<sup>17</sup>, AIHWKIT exposed analog HAT and inference simulation in a practical mixed-signal software stack<sup>18</sup>, and CrossSim provides a GPU-accelerated crossbar-accuracy workflow with parasitic resistance, ADC, drift, and lookup-table-based training models<sup>19</sup>. These mature simulators were developed primarily around inorganic resistive memories, and they do not directly expose inverse-gamma optoelectronic photoresponse or organic-specific double-exponential retention as first-class device primitives. Our behavioral profile-driven approach therefore serves as an organic-specific complement rather than as a direct replacement for these inorganic toolkits.

In the CIM context, dense linear operators are natural crossbar targets, whereas depthwise and highly irregular operators often suffer from poor utilization<sup>37–39</sup>. Recent heterogeneous CIM accelerators for Vision Transformers confirm this pattern: linear projections (query-key-value, or QKV, output, feed-forward) are mapped to analog arrays, whereas attention scores, softmax normalization, and other dynamic interactions usually remain digital or are delegated to specialized mixed-signal subpaths because of their precision requirements and irregular access patterns<sup>27;28</sup>. Analog memristor accelerators for self-attention have also been explored as a competing direction<sup>40–42</sup>, but they rely on a substantially different hardware partitioning strategy from the conservative hybrid mapping adopted here. This separation underlies our



**Figure 1:** Cross-dataset accuracy under canonical deployment (4-bit quantization, 5% C2C, 10% D2D variability). Error bars denote  $\pm 1$  standard deviation where Monte Carlo (MC) statistics are available; bars without visible error bars indicate deterministic baselines or currently available point estimates.

mapping policy. Query-key-value projections, output projections, feed-forward layers, and patch-embedding convolutions are treated as analog candidates, while depthwise convolutions, normalization, and dynamic attention products remain digital. We adopt this hybrid logic, extending prior CIM simulators to organic-device profiles and Tiny-ViT operators.

Unlike prior ViT-on-PIM studies, we jointly model photoresponse, profile substitution, variability, retention, ADC resolution, and front-end distortion within one workflow. This focus converges with recent low-bit ViT quantization findings, which repeatedly identify attention-adjacent computations as especially sensitive under aggressive precision reduction<sup>29–31</sup>. Extending established simulator practice to organic optoelectronic domains, the present framework links reported device characteristics to CIFAR-scale transformer benchmarks through a replaceable profile interface.

## 3 Results

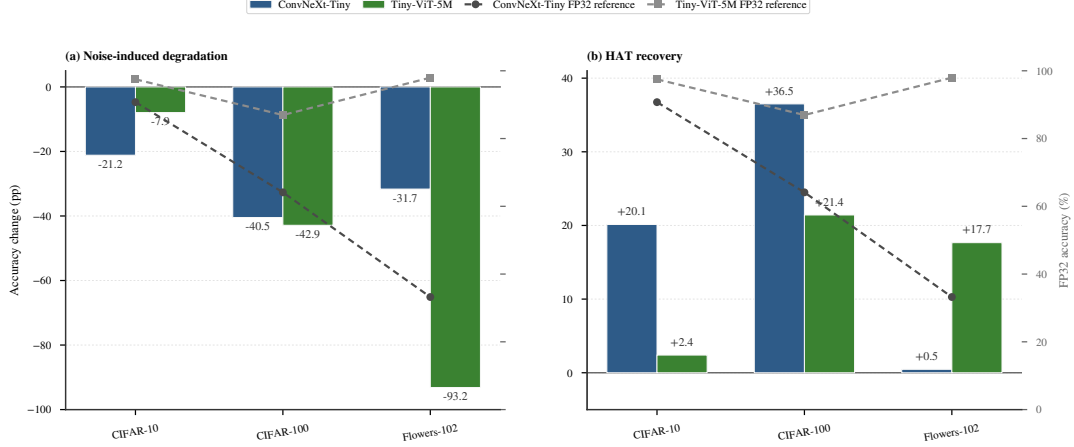
### 3.1 Baseline Digital Performance

Digital FP32 baselines are 95.46% (ResNet-18), 90.74% (ConvNeXt-Tiny), and 98.06% (Tiny-ViT-5M) on CIFAR-10, with corresponding CIFAR-100 values of 78.64%, 64.12%, and 86.94% as listed in the Supplementary Information. All accuracy values reported for noisy and HAT deployments are best-checkpoint results unless otherwise stated. ConvNeXt probes analog adaptation from random initialization, whereas Tiny-ViT probes foundation-model deployment.

### 3.2 Quantization and Noise Resilience

Quantization alone introduces only a modest penalty in 4-bit hybrid models. The zero-noise hybrid control (V2; Table 1) retains  $97.39 \pm 0.00\%$  ( $n = 10$  deterministic evaluations), suggesting that 4-bit mapping by itself is not the dominant source of error in the present setup. Under the canonical uniform-noise profile, the recovered-weight law later formalized in Eq. 1 suppresses much of the effective analog perturbation below the 4-bit least significant bit, so many noisy conductance realizations are mapped back into the same recovered weight bin after calibration. This behavior disappears once the noise law becomes state-dependent (Section 3.7).

Deployment fragility increases with task complexity (Figs. 1–2). On CIFAR-100, HAT improves Tiny-ViT from 44.06% to 65.48% and ConvNeXt from 23.86% to 60.54%. Flowers-102 shows limited recovery (Supplementary Table S7), but the zero-noise hybrid control reaches 91.30%, confirming that failure is driven by noise-data interaction rather than hybrid deployment alone.



**Figure 2:** Accuracy degradation from FP32 to noisy deployment and HAT recovery. Tiny-ViT: 10-run MC; ConvNeXt: single-run estimates shown.

An extended ADC sweep confirms the same 6-bit threshold on ConvNeXt-Tiny (Supplementary Fig. S3). The iso-accuracy contour analysis (Section 3.6) shows that this cliff is largely independent of device variability, producing a consistent  $\sim 7$  pp transition across all tested  $\sigma_{D2D}$  levels.

### 3.3 Retention and Temporal Drift

The corrected retention-aware follow-up model (V8) exhibits a rapid early drop (91.63% at 0 s  $\rightarrow$  82.66% at 1 s) followed by a broad plateau near 79% (79.13–79.51%) through 10 000 s (Supplementary Fig. S6), consistent with the uniform decay model being adequate in the present regime.

### 3.4 Hardware-Instance Transferability

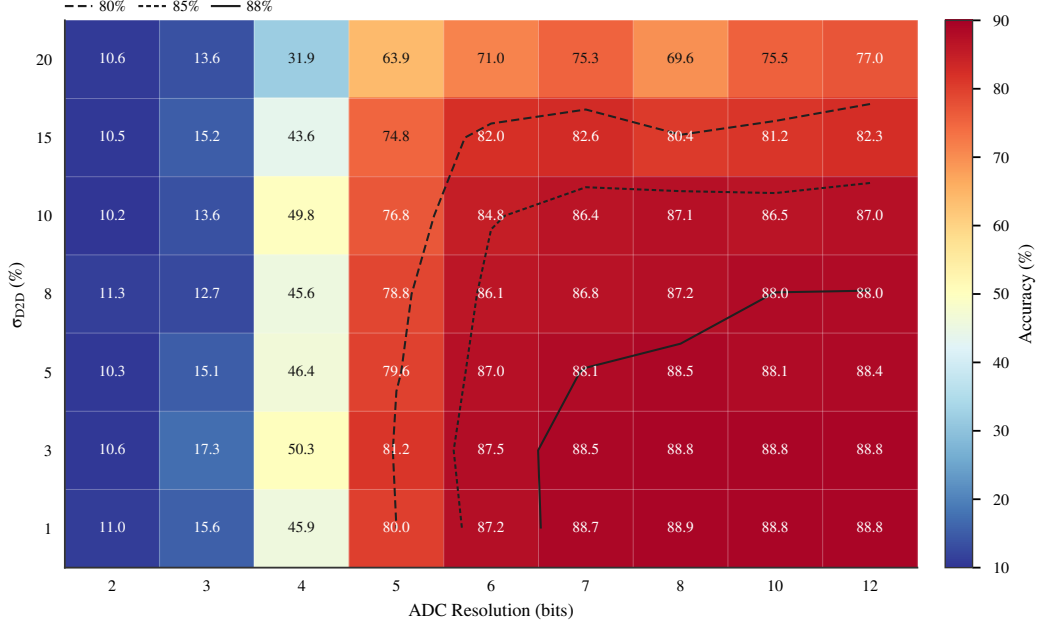
Fresh-instance evaluation reveals that standard fixed-mask HAT (V4) collapses to chance level (10.00%) on arrays with different fixed D2D realizations, indicating pronounced hardware-instance overfitting. In this setting the 10.00% result reflects a collapsed single-class predictor on class-balanced CIFAR-10 rather than a noisy dispersion around chance. The collapse was independently confirmed under FP32 inference with autocast disabled across the same 10 fresh D2D instances and 5 Monte Carlo evaluations per instance, yielding  $10.00 \pm 0.00\%$  in the no-AMP recovery run (`fresh_instance_eval_v4_standard_noamp.json`). The single-class predictor is a deterministic outcome of fixed-mask training under epoch-resampled D2D, not a numerical artifact. This motivates the distribution-matched Ensemble HAT objective (Eq. 3), formally defined in Section 5 and instantiated in Section 3.7.

### 3.5 Physical Front-end Compensation

The organic phototransistor front end exhibits a sublinear photoresponse  $I_{\text{photo}} \propto P^{\gamma_{\text{phys}}}$  that we compensate with an inverse-gamma preprocessor  $P_{\text{in}} = X^{1/\gamma_{\text{phys}}}$  (Eq. 5). A systematic  $\gamma_{\text{phys}}$  scan in the Supplementary Information shows that compensation restores **+5.8 pp** at  $\gamma_{\text{phys}} = 2.0$  (89.85% vs. 84.04%, single-seed deterministic evaluation), but amplifies shot-noise variance for  $\gamma_{\text{phys}} < 1$ . The trade-off is captured analytically in Supplementary Figs. S7–S8.

### 3.6 Iso-Accuracy Operating Envelope

A 63-point joint sweep over  $\sigma_{D2D} \in \{1, 3, 5, 8, 10, 15, 20\}\%$  and  $\text{ADC} \in \{2, 3, 4, 5, 6, 7, 8, 10, 12\}$  bits (10 MC runs per point) reveals three regimes (Figure 3). Below 5-bit ADC, accuracy



**Figure 3: Iso-accuracy contour map** for the Ensemble HAT Tiny-ViT model under joint  $\sigma_{D2D}$  and ADC precision sweep ( $\sigma_{C2C} = 5\%$ ,  $NL = 1.0$ , 10 MC runs per point). The 6-bit ADC cliff and the D2D-dominated degradation in the operational regime are visible.

collapses to near-chance; the 5-to-6-bit transition yields a consistent  $\sim 7$  pp jump (mean across all seven D2D levels; the per-row jump ranged from 5.8 to 8.1 pp). Above 6 bits, degradation is dominated by D2D:  $\leq 10\%$  D2D maintains  $> 84\%$ , while  $20\%$  D2D reduces it to  $71\text{--}77\%$ . Sobol analysis (Eq. 8; defined in Section 5) confirms the two-phase structure:  $S_{ADC} = 0.98$  over the full grid, and  $S_{D2D} = 0.92$  in the operational region ( $\geq 6$  bits,  $\sigma_{D2D} \leq 15\%$ ), with negligible interaction ( $< 4\%$ ).

### 3.7 Nonlinear Writing and Hardware-Aware Training

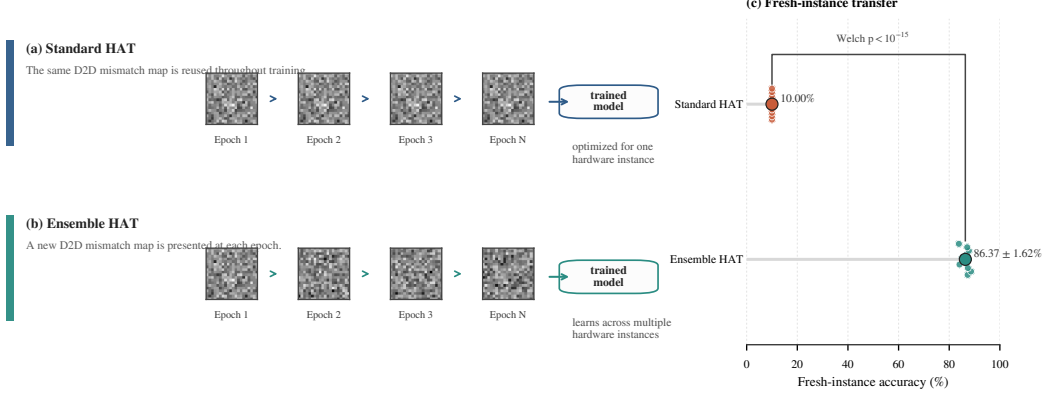
Proportional-noise HAT recovers Tiny-ViT to  $97.37 \pm 0.05\%$ , whereas the baseline severe-nonlinearity configuration ( $NL = 2.0$ ) recovers only  $27.72 \pm 0.82\%$  under the present gradient-scaling recipe. Ensemble HAT (Figure 4; formally defined in Section 5, Eq. 3) preserves  $86.37 \pm 1.54\%$  across 10 fresh arrays, overwhelming standard HAT (one-sample  $t$ -test against the chance-level baseline,  $n = 10$  fresh arrays per arm,  $p < 10^{-15}$ ). An exploratory single-run training-ablation cadence scan (Supplementary Fig. S4) shows epoch-level resampling reaches  $88.41\%$  (50-epoch training ablation), versus  $87.18\%$  (fixed) and  $86.16\%$  (per-batch), confirming that structured epoch-level resampling rather than per-batch perturbation is the load-bearing schedule choice.

The full accuracy matrix is provided in the Supplementary Information. Tiny-ViT demonstrates superior baseline accuracy and significant recovery under HAT, especially on complex tasks, though it remains sensitive to nonlinear write dynamics.

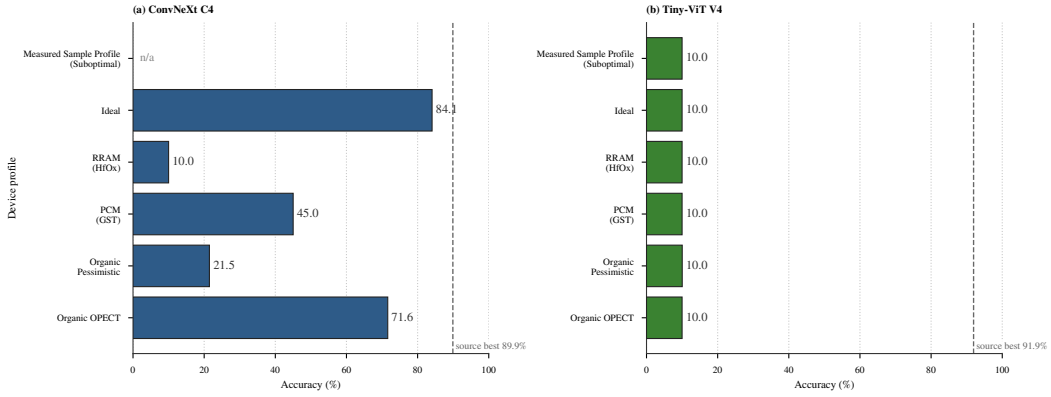
To test whether the ceiling is structural, we trained a joint MLP-linear + Ensemble HAT model under  $NL = 2.0$ . The checkpoint reaches  $91.36\%$  on the source training instance, but fresh-instance evaluation across 10 arrays yields only  **$30.53 \pm 7.07\%$** . This falsifies the hypothesis that coupling severe-NL compensation with distributional resampling breaks the severe-NL ceiling; the  $\sim 30\%$  barrier persists.

The convergence of three independent mitigations (MLP-only, all-linear, joint) on the same  $\sim 30\%$  fresh-instance ceiling under  $NL = 2.0$  suggests that severe write nonlinearity introduces a generalization barrier in the attention pathway that training-recipe modifications cannot overcome within the first-order surrogate regime.





**Figure 4: Ensemble HAT concept.** Standard HAT reuses one fixed D2D mismatch map throughout training, whereas Ensemble HAT resamples the mismatch map each epoch. The fresh-instance transfer panel highlights the resulting generalization gap between the two training protocols.



**Figure 5: Zero-shot transfer across alternative device profiles.** Bars report the accuracy of standard-HAT checkpoints when evaluated on literature-calibrated or measured alternative device profiles without profile-specific retraining. ConvNeXt C4 retains partial transfer on the OPECT and idealized profiles but degrades sharply on phase-change memory (PCM) and resistive random-access memory (RRAM), whereas Tiny-ViT V4 collapses to chance level (10.00%) across all shown alternatives. Dashed vertical lines mark the source-profile best accuracy, and unavailable architecture/profile combinations are labeled as n/a.

### 3.8 Case Study: Zero-Shot Transfer to a Literature-Calibrated Device

As a literature-anchored reference point, we simulated zero-shot transfer to a 2025 OPECT array<sup>10</sup> ( $\sigma_{C2C} = 2\%$ ,  $\sigma_{D2D} = 3\%$ ). Figure 5 shows that the standard-HAT checkpoints transfer poorly to these alternative profiles: ConvNeXt retains only partial transfer, whereas Tiny-ViT V4 collapses to chance level (10.00%) across all shown alternatives. By contrast, the Ensemble-HAT checkpoint reaches **88.53 ± 0.08%** ( $n = 10$  fresh-instance evaluations) under the nominal OPECT proxy in the same workflow.

This literature-anchored case study illustrates how the framework links reported physical metrics to algorithm deployment risk. The stark contrast between standard and Ensemble HAT on literature-calibrated benchmarks suggests that addressing spatial mismatch map overfitting is important for realizing the potential of emerging organic optoelectronic arrays.

## 4 Discussion

### 4.1 Principal Accuracy Bottlenecks

The combined ResNet, ConvNeXt, and Tiny-ViT analyses indicate that the dominant limitations under the canonical deployment regime (standard 4-bit quantization with 5% C2C and 10% D2D variability) are not the ones most often emphasized at the device level. Once hardware-aware training (HAT) and scale recovery are available, standard cycle-to-cycle (C2C) and device-to-device (D2D) variability do not dominate the error budget. Three constraints dominate instead.

A quantitative decomposition supports this ordering. Using the grid-based variance-decomposition definition in Eq. 8, the  $7 \times 9$  D2D–ADC grid shows that, over the full parameter space, ADC precision accounts for 97.6% of accuracy variance ( $S_{\text{ADC}} = 0.98$ ), while D2D contributes only 1.8%. However, conditioning on the operational envelope ( $\geq 6$ -bit ADC,  $\sigma_{\text{D2D}} \leq 15\%$ ), the hierarchy inverts: D2D variability accounts for 92.2% of residual variance ( $S_{\text{D2D}} = 0.92$ ). This two-phase structure has a direct hardware implication: designers should first secure 6-bit readout, then invest the remaining error budget in reducing device-to-device mismatch.

The first is ADC resolution. The transition around 6-bit ADC resolution marks a clear change in model behavior: below this point transformer inference degrades substantially, indicating that improvements in conductance control remain gated by readout precision in the present mixed-signal stack.

The second is hardware-instance overfitting. The collapse of V4 under fresh-instance transfer (10.00% accuracy) shows that standard HAT readily fits a single fixed D2D realization. Replacing the fixed-mask objective in Eq. 2 with the mismatch-distributed objective in Eq. 3 raises fresh-instance accuracy to  $86.37 \pm 1.54\%$ , indicating that multi-instance-aware training can mitigate this transfer failure mode. Mechanistically, Ensemble HAT changes the fixed spatial mismatch map itself across epochs rather than merely perturbing one nominal instance.

A residual gap to the 98.06% digital baseline persists under matched deployment conditions. Part of this difference may reflect regularization from per-epoch D2D resampling rather than analog noise alone. Capacity-aware Ensemble HAT schedules remain an open optimization problem.

The third is the scale-masking regime observed in V2. The high accuracy of V2 under standard uniform noise is a conditional property of the canonical simulator configuration rather than a universal form of robustness. When the recovered-weight law in Eq. 1 maps analog perturbations below the 4-bit quantization step, many realizations remain in the same recovered weight bin after calibration. This protection depends on accurate post-array gain calibration and disappears once the noise law becomes state-dependent, as shown by the proportional-noise experiments.

### 4.2 Transformer Sensitivity to Non-Idealities

Comparing the dual testbeds—ConvNeXt (CNN) and Tiny-ViT (Transformer)—reveals that the Transformer architecture is more fragile under the present physical stress tests. Because the two models serve complementary roles (ConvNeXt from scratch versus Tiny-ViT from pre-trained fine-tuning), this result should be read as evidence of a strong architecture-and-training interaction rather than as a universal ranking of all Transformers against all CNNs.

The proportional-noise experiments illustrate this most clearly. The collapse of V4 under proportional noise (10.00% accuracy), contrasted with the relative stability of ConvNeXt under moderate nonlinearities, indicates that Transformer inference depends on the precision of token projections. Because attention maps arise from global interactions, perturbations at a small number of high-conductance devices can disrupt the spatial weighting of the entire block.



**Why the transformer amplifies front-end distortion.** The larger accuracy drop in V6 than in ResNet R4 reflects a structural difference between the two architectures. In a CNN, local convolution and pooling average perturbations spatially, and ReLU saturation limits the propagation of intensity-dependent scaling errors. In a ViT, patch embeddings are fed directly into global self-attention: a sublinear photoresponse alters the token embedding, and the softmax exponentiates similarity changes, redistributing attention weights across the entire sequence without spatial averaging. The detailed mechanism and quantitative support are discussed in the front-end theory note of the Supplementary Information.

Robustness is highly regime-matched. Tiny-ViT can recover strongly when trained and evaluated under the same proportional-noise law, yet the resulting checkpoint does not transfer back to the canonical uniform-noise regime. In this framework, robustness is distribution-specific rather than noise-invariant.

The framework remains reasonably resilient overall. A single-run compound stress test combining all modeled non-idealities (2% C2C, 3% D2D, 6-bit ADC, 1% IR drop, 1% sneak path, and 1,000 s retention) maintained an 89.61% accuracy for Tiny-ViT, suggesting that the mixed-precision architecture with Ensemble HAT can tolerate moderate cumulative deployment stress within the present model.

### 4.3 Task Complexity and Data Starvation

HAT recovery is most valuable on CIFAR-100, where decision boundaries are fine, whereas CIFAR-10 is simple enough that noisy standard models remain functional. Flowers-102 marks the low-data extreme (Supplementary Table S7).

### 4.4 The "Analog Ceiling" and Energy Efficiency

A first-order energy model projects roughly an order-of-magnitude dense-projection energy reduction versus FP32 (approximately  $11\times$ ), reduced to approximately  $10\times$  after 10–50% routing overhead. Digital attention still dominates total energy (roughly 60%). These are system-level upper bounds under placeholder constants, not chip-predictive estimates.

### 4.5 Limitations

Energy parameters are placeholders; parasitic ranges derive from ReRAM literature<sup>43;44</sup> until measured organic-array data become available. Three circuit-level phenomena are not yet modeled explicitly: spatial IR drop (currently a 1% scalar placeholder), sneak-path currents (likewise), and temperature-dependent shifts in photoresponse and mismatch variance. Heavy-tailed conductance distributions are also absent. Spatially correlated D2D is evaluated in Supplementary Note SX.Z: under separable AR(1) perturbation, moderate correlation ( $\rho = 0.3$ ) degrades Ensemble HAT fresh-instance accuracy by 1.76 pp versus the matched i.i.d. baseline ( $84.57 \pm 2.39\%$  vs.  $86.33 \pm 1.61\%$ ), and stronger correlation ( $\rho = 0.5$ ) degrades it by 4.20 pp ( $82.12 \pm 3.95\%$ ). Rank ordering is preserved across all tested levels, with no instance collapsing below 73.7% (`fresh_instance_eval_v4_ensemble_correlated_d2d.json`). Scale-masking depends on ideal calibrated rescaling; a hook-based check shows that moderate ADC calibration errors change V4 accuracy by only 0.18 pp. The energy model further assumes ideal single-shot programming and therefore excludes iterative write-verify overhead for 4-bit conductance states. The  $NL = 2.0$  limit reflects the present gradient-scaling surrogate, not a material bound. Group-wise ablation confirms that the bottleneck is concentrated in the MLP analog path, while both attention-side linearizations (QKV and projection) collapse structurally. Validation against fabricated organic arrays is deferred to future work. Overall, the framework ranks deployment risks, not a chip-predictive emulator.

## 4.6 Outlook

Next steps include automated profile fitting from laboratory characterization and extended multi-instance HAT schedules. Hardware-in-the-loop validation—correlating simulator predictions against measured organic array conductance distributions and task-level accuracy—is the highest-priority next step; the present results are intentionally a simulation baseline under literature priors, preserving a clean pathway for later recalibration without code changes. The natural extension of the present behavioral framework is a circuit-aware layer that models spatial IR drop, sneak paths, and temperature-dependent coefficients from array geometry—a scope we have explicitly deferred here to prioritize materials-to-system risk ranking before full hardware closure. Extrapolation to ImageNet-scale deployment is also outside the present evidence base: per-epoch D2D resampling would incur substantially higher training overhead, the 6-bit ADC cliff may shift upward for deeper MLP stacks with finer decision boundaries, and the current results do not address fresh large-scale training from random initialization. A preliminary cross-framework comparison against CrossSim on a deterministic 1 000-image CIFAR-10 test subset (single-run clean baseline, 3-run Monte Carlo under noise) shows consistent baseline inference (86.2% ours vs. 83.7% CrossSim at 8-bit ADC) but diverges under noise injection ( $81.63 \pm 0.56\%$  vs.  $67.20 \pm 2.67\%$  at  $\sigma = 5\%$ , a large qualitative divergence of 14.43 pp at  $n = 3$ , preliminary), highlighting the sensitivity of accuracy predictions to the noise-to-conductance mapping. The subset protocol is disclosed explicitly because of CrossSim throughput constraints; see Supplementary Note SX.Y.

## 5 Methodology

### 5.1 Hybrid Analog/Digital Deployment

We adopt a hybrid analog/digital inference stack: dense linear operators execute on differential-pair crossbars; control-heavy operators remain digital. Concretely, the analog partition contains patch embeddings, query/key/value (Q/K/V) and output projections, and feed-forward layers; the digital partition contains softmax, layer normalization, GELU activation, positional encoding, residual additions, and the class token. The system-level split and the operator-level rationale are summarized in the Supplementary Information following established practice in heterogeneous CIM accelerators<sup>27;28;37–39</sup>.

Under this mapping, approximately 88% of parameters are assigned to analog execution, whereas roughly 60% of the estimated energy remains in the digital domain under the present placeholder constants, largely because of attention operations. This split reflects a basic architectural constraint: analog crossbars are well suited to weight-stationary matrix multiplication, but much less suited to dynamic, input-dependent computations<sup>27</sup>. Recent heterogeneous CIM accelerators for Vision Transformers follow the same logic, keeping attention digital even as linear projections move to analog arrays<sup>28;38;39</sup>. The present organic device literature likewise demonstrates regular static array programming more naturally than fully dynamic interaction engines<sup>9;10</sup>. Layer-wise mapping ablations that vary the analog–digital boundary are supported by the framework’s modular operator assignment, but they are deferred here so that the present study can isolate the impact of device non-idealities on a single, widely adopted hybrid partition.

### 5.2 Modeling Physical Non-Idealities

The framework injects quantization, cycle-to-cycle (C2C) and device-to-device (D2D) variability, ADC distortion, retention drift, and nonlinear-write surrogates directly into the forward path. Each analog-mapped weight is decomposed into positive and negative branches, mapped to the conductance window  $[G_{\min}, G_{\max}]$ , and quantized to  $n_{\text{states}}$  levels via a straight-through estimator (STE) quantizer  $Q_n(\cdot)$ . Differential readout recovers  $G_{\text{eff}} = \tilde{G}^+ - \tilde{G}^-$ , and the downstream

digital weight is

$$\hat{W}_{ij} = s_\ell G_{\text{eff},ij}, \quad s_\ell = \begin{cases} \frac{w_{\text{max}}}{G_{\text{max}} - G_{\text{min}}}, & \text{standard calibration} \\ \frac{w_{\text{max}}}{\max_{i,j} |G_{\text{eff},ij}| + \epsilon}, & \text{retention-time recalibration} \end{cases} \quad (1)$$

with layer index  $\ell$  omitted. Unless otherwise stated, we use the standard-calibration branch; the retention-recalibration branch is used only for the V8 retention-drift experiments (Section 3.3). The recovered weight step is  $\Delta_{\hat{W}} = s_\ell(G_{\text{max}} - G_{\text{min}})/(n_{\text{states}} - 1)$ , so perturbations smaller than half a step are masked. The differential-pair scheme suppresses common-mode noise<sup>45;46</sup>.

Hardware-aware training (HAT) keeps forward perturbations active during optimization while back-propagating through a straight-through estimator. Standard HAT fixes one D2D mismatch field  $M_0$ ,

$$\theta_{\text{std}}^* = \arg \min_{\theta} \frac{1}{N} \sum_{n=1}^N \ell(f(x_n; \theta, M_0), y_n), \quad (2)$$

whereas Ensemble HAT resamples  $M^{(e)} \sim p(M)$  at each epoch,

$$\theta_{\text{ens}}^* = \arg \min_{\theta} \mathbb{E}_{M \sim p(M)} \left[ \frac{1}{N} \sum_{n=1}^N \ell(f(x_n; \theta, M), y_n) \right]. \quad (3)$$

The expectation is Monte Carlo–estimated with one freshly sampled mismatch map per training epoch, held constant across all mini-batches within that epoch. Fresh-instance transfer evaluates on unseen draws  $M' \sim p(M)$ :

$$\text{Acc}_{\text{fresh}} = \mathbb{E}_{M' \sim p(M)} [\text{Acc}(f(\cdot; \theta_{\text{ens}}^*, M'))]. \quad (4)$$

Each fresh array corresponds to an i.i.d. Gaussian draw  $M' \sim \mathcal{N}(0, \sigma_{\text{D2D}}^2)$  held constant over the evaluation pass, distinct from the cycle-to-cycle perturbation that re-samples per forward pass. For the canonical Ensemble HAT transfer result, each fresh-instance mean is itself the mean of five forward-pass Monte Carlo evaluations, and the reported  $\pm 1.54$  pp spread is the standard deviation across the ten fresh-instance means rather than across all fifty forward passes pooled together.

### 5.3 Training Protocol

All Tiny-ViT experiments are trained with AdamW ( $lr = 5 \times 10^{-4}$ , weight decay 0.05) for 100 epochs using a batch size of 64 and a cosine annealing learning-rate schedule ( $T_{\text{max}} = 100$ ). Images are resized to  $224 \times 224$ , augmented with random horizontal flips, and normalized by dataset statistics. The default analog programming resolution is  $n_{\text{states}} = 16$  conductance levels. Experiment V1 uses standard digital training; V2 uses hybrid layers without noise; V3 uses hybrid layers with fixed D2D mismatch ( $\sigma_{\text{D2D}} = 10\%$ ) active during training but no per-forward C2C resampling; V4–V7 employ full hardware-aware training in which fixed D2D mismatch is resampled at the start of every epoch and C2C noise ( $\sigma_{\text{C2C}} = 5\%$  or  $10\%$ ) is drawn on each forward pass. V6 additionally enables the physical photoresponse front end, and V7 evaluates post-training retention decay. Complete experiment settings and the open-source training scaffold are summarized in the Supplementary Information and repository release package.

First-order nonlinear-write and double-exponential retention surrogates are supported, with parameters anchored to measured dinaphtho-thieno-thiophene (DN TT) transients<sup>47</sup>. Nonlinear write modifies the STE backward pass by scaling the straight-through gradient with a state-dependent factor proportional to the present conductance magnitude raised to  $NL - 1$  (Supplementary Eq. S2 gives the exact piecewise form); ideal write is recovered when  $NL_{\text{LTP}} = 1$  (long-term potentiation) and  $|NL_{\text{LTD}}| = 1$  (long-term depression). For optoelectronic sensing,

we incorporate a physical front-end model (V6) that first applies inverse-gamma preprocessing to the normalized input  $X \in [0, 1]$ ,

$$P_{\text{in}} = X^{1/\gamma_{\text{phys}}}, \quad (5)$$

and then propagates it through a photocurrent response model,

$$I_{\text{photo}} = \alpha P_{\text{in}}^{\gamma_{\text{phys}}} + I_{\text{dark}} + \varepsilon_{\text{shot}} \approx \alpha X + I_{\text{dark}} + \varepsilon_{\text{shot}}, \quad (6)$$

with  $\text{Var}[\varepsilon_{\text{shot}}] \propto I_{\text{photo}}$ . The proportionality coefficient is absorbed into the per-profile front-end calibration, so its exact value (governed by photodetector quantum efficiency and integration time) is implicit in the reported results. The uncompensated form ( $P_{\text{in}} = X$ ) and the front-end theory derivation are given in the Supplementary Information. The simulated photocurrent is finally renormalized on a per-sample basis,

$$X_{\text{norm}} = \frac{I_{\text{photo}} - I_{\text{min}}}{I_{\text{max}} - I_{\text{min}} + \epsilon}, \quad (7)$$

before dataset normalization. The released profiler uses representative edge-node energy and latency parameters.

The full weight-to-conductance mapping pipeline is illustrated in the Supplementary Information.

## 5.4 Profile-Driven Substitution Interface

The profile interface packages technology-specific assumptions into a replaceable JSON parameter bundle ( $\sigma_{\text{D2D}}, \sigma_{\text{C2C}}, n_{\text{states}}, G_{\text{min}}, G_{\text{max}}, NL_{\text{LTP}}, NL_{\text{LTD}}, \tau_1, \tau_2, A_0, \gamma_{\text{phys}}, I_{\text{dark}}, \dots$ ). The deployment evaluation holds the digital backbone and training recipe fixed while substituting the profile into the same mixed-signal inference path, enabling literature priors today and measured-device characterizations later without code changes. Representative profile fields and provenance are documented in the Supplementary Information.

## 5.5 Sensitivity and Energy Metrics

For the joint D2D–ADC sweep, we quantify first-order factor importance with Sobol indices

$$S_i = \frac{\text{Var}_{X_i}(\mathbb{E}[Y | X_i])}{\text{Var}(Y)}, \quad (8)$$

estimated directly from the  $7 \times 9$  grid of Monte Carlo means as  $\text{Var}(\text{mean}_{\text{grid}}[X_i])/\text{Var}(Y)$ , with MC noise realizations drawn independently per grid point and per architecture (seeds are not shared across the grid or across models). Indices are reported over both the full grid and the operational subregion. Energy is estimated via a first-order analytical model described in the Supplementary Methods from operator counts and placeholder per-operation constants; the numbers are intended for relative comparison rather than routed-circuit prediction.

# 6 Experimental Setup

We evaluate ResNet-18 on CIFAR-10 as an entry validation platform, ConvNeXt-Tiny trained from scratch as the higher-capacity CNN testbed, and Tiny-ViT-5M fine-tuned from ImageNet as the deployment-oriented transformer setting. CIFAR-10, CIFAR-100, and Flowers-102 provide a progression from easier to harder classification settings and from relatively data-rich to data-limited regimes.

The experiments fall into three groups: ResNet baseline quantization and front-end trends, ConvNeXt retention and proportional-noise studies, and Tiny-ViT hardware-aware training

**Table 1:** Notation used for the Tiny-ViT experiment family in the main text.

| ID | Meaning                                                                             |
|----|-------------------------------------------------------------------------------------|
| V1 | FP32 digital baseline                                                               |
| V2 | Quantized hybrid control without analog noise                                       |
| V3 | Standard training with a fixed D2D realization; noisy deployment baseline           |
| V4 | Uniform-noise hardware-aware training result used as the canonical deployment model |
| V5 | Pessimistic robustness stress test with larger uniform noise                        |
| V6 | V4 plus physical front-end compensation                                             |
| V7 | Legacy retention-aware model, excluded from the canonical claims                    |
| V8 | Corrected retention-aware follow-up used only for retention checks                  |

together with physical-extension tests such as nonlinear write and fresh-instance transfer. Detailed optimization settings and the Monte Carlo evaluation protocol are summarized in the Supplementary Information.

For ConvNeXt, C1–C4 denote the analogous FP32, quantized/no-noise, noisy-baseline, and hardware-aware training runs; C4 also anchors the proportional-noise and retention follow-up studies described in the Supplementary Information.

Parameter provenance is summarized in the Supplementary Information, together with the optimization settings and evaluation conditions used for the reported runs. The present study therefore targets reproducible reruns through archived configurations, logs, and model lineage. We also emphasize that the framework operates at the simulation level: all device parameters are literature-derived or proxy-calibrated rather than extracted from fabricated devices under direct measurement. Validation against physical hardware remains for future work.

## 7 Conclusion

We presented a first-order behavioral simulation framework for organic optoelectronic compute-in-memory inference that links materials-level characterization to task-level vision performance. By combining differential mapping, variability, retention, and peripheral constraints within a profile-driven workflow, the framework serves as a practical materials-to-system decision aid for evaluating deployment risk before full array hardware becomes available.

Several observations stand out. A 63-point iso-accuracy sweep reveals a two-phase constraint hierarchy: ADC resolution below 6 bits causes abrupt collapse ( $S_{\text{ADC}} = 0.98$  over the full grid), while within the operational envelope, device-to-device variability becomes the binding factor ( $S_{\text{D2D}} = 0.92$ ). Fixed hardware-instance transfer, rather than nominal quantization, is the most prominent simulated deployment risk in the present regime: standard HAT collapses on fresh arrays, whereas Ensemble HAT raises fresh-instance accuracy from 10.00 % to  $86.37 \pm 1.54\%$ . In the canonical uniform-noise setting, quantization and nominal stochasticity are less limiting once scale recovery is active. By contrast, the baseline severe write nonlinearity setting remains difficult to recover under the present gradient-scaling approximation ( $27.72 \pm 0.82\%$ ), and determining whether physical organic devices enter this regime will require direct nonlinear-write measurements.

The framework translates reported device behavior into deployment-facing system risk through an explicit profile interface. Under a literature-anchored OPECT case study, zero-shot simulated deployment reaches  $88.53 \pm 0.08\%$  ( $n = 10$  fresh-instance evaluations) without any change to the evaluation workflow. The same structure should simplify future measured-device calibration while preserving model-level comparability within a common workflow.

## Data Availability

All datasets used in this study (CIFAR-10, CIFAR-100, and Flowers-102) are publicly available through standard PyTorch torchvision and HuggingFace dataset interfaces. The source

data underlying the main-text and Supplementary figures and tables are packaged in the submission-matched source-data archive together with the corresponding manifests, logs, and figure-generation inputs. A reproducibility archive containing the code snapshot, source data, SHA-256 manifest, and CITATION.cff file has been staged for Zenodo deposition and will be assigned a DOI at acceptance.

## Code Availability

The simulation framework, training and evaluation scripts, device-profile utilities, and plotting code will be provided to editors and reviewers through a reviewer-accessible archive at submission. A curated public release will be made available at [https://github.com/OrgOptEdge/compute\\_vit](https://github.com/OrgOptEdge/compute_vit) upon acceptance. The current repository is distributed under the Apache-2.0 licence, and the staged Zenodo archive will be updated with the final public DOI at acceptance.

## Competing Interests

The authors declare no competing interests.

## Author Contributions

S.L. conceived the framework, implemented the simulator, conducted all experiments, and wrote the manuscript.

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